Predicting bearing degradation from run-to-failure data via supervised learning

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Rolling element bearing degradation is a primary cause of unexpected downtime in industrial machinery. This study proposes a supervised machine learning framework for multi-state bearing health classification using time-domain vibration telemetry from the NASA IMS Dataset 1. Statistical features—Standard Deviation, RMS, Maximum Peak, and Kurtosis—were extracted across an 8-channel sensor array, while Mean was excluded due to its lack of diagnostic sensitivity. A sensor-specific partitioning scheme categorized the operational lifespan into Normal (Class 1), Warning (Class 2), and Critical (Class 3) states. To resolve severe class imbalance, an ensemble RUSBoosted Trees model was trained using 5-fold cross-validation and a custom misclassification cost matrix. The framework achieved class-balanced True Positive Rates (TPR) of 91.0% (Normal), 83.6% (Warning), and 95.6% (Critical). Crucially, it eliminated high-risk critical-to-normal false negatives (0.0% error rate), demonstrating high reliability for safety-critical predictive maintenance systems

Keywords: Rolling element bearing, Fault diagnosis, Time-domain feature extraction, Class imbalance, RUSBoosted trees, Predictive maintenance, NASA IMS dataset.

I. INTRODUCTION

Rolling element bearings are critical components in rotating machinery. Progressive degradation over time is inevitable under continuous mechanical stress, often leading to catastrophic failure and costly downtime if undetected. Consequently, Condition Monitoring (CM) and Predictive Maintenance (PdM) techniques have become essential for industrial asset management. While traditional methods rely on fixed thresholding or manual inspection, machine learning (ML)—particularly supervised classification—offers a promising data-driven alternative for automated fault diagnosis using historical run-to-failure vibration data.

However, real-world degradation prediction faces major technical hurdles: high-frequency noise in non-stationary signals, the lack of prior labels for dynamic health state transitions, and severe class imbalance, as normal operating data heavily outnumbers degraded states.

To address these challenges, this study proposes a supervised classification framework for multi-class bearing degradation prediction (Normal, Warning, and Critical) using run-to-failure vibration datasets from the NASA IMS test center. Key statistical features were extracted across sensor channels and dynamically labeled via automated moving-mean thresholding. Imbalance mitigation techniques, including misclassification cost matrices and ensemble methods (e.g., RUSBoosted and Bagged Trees), were evaluated to optimize early fault detection. The primary objective is to establish an effective predictive pipeline that balances classification accuracy and True Positive Rates (TPR) across all health stages, ensuring reliable safety.

II. EXPERIMENTAL SETUP

The experimental validation of this study utilizes the benchmark bearing run-to-failure dataset provided by the Center for Intelligent Maintenance Systems (IMS), University of Cincinnati. Specifically, Dataset 1 is selected to evaluate the proposed supervised classification framework.

As illustrated in Fig. 1, the experimental test rig consists of an AC motor driving a shaft via a mechanical rubber belt system, maintaining a constant rotational speed of 2,000 RPM under a radial load of 6,000 lbs. applied to the shaft and bearings. Four Rexnord ZA-211 double-row roller bearings are installed in series along the rotating shaft. High-sensitivity accelerometers are mounted on each bearing housing to capture multi-channel vibration signals, while thermocouples are embedded to record operating temperatures, as schematized in Fig. 2.

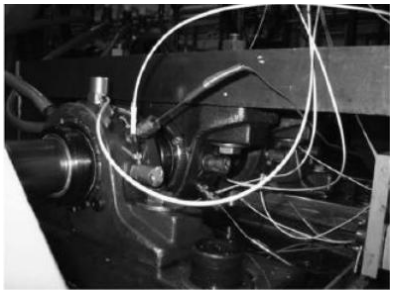


Fig. 1 Actual experimental test rig setup showing the motor drive, shaft assembly, and instrumented bearing housings.

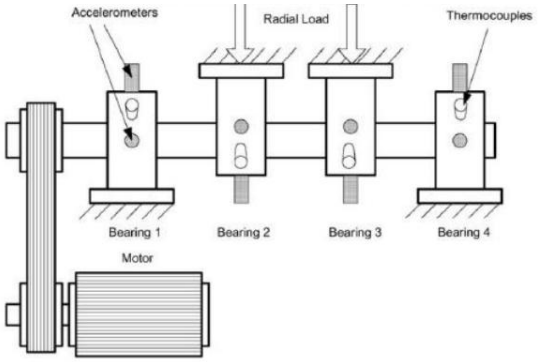


Fig. 2 Schematic diagram of the NASA IMS bearing test rig detailing the positioning of accelerometers, thermocouples, and the applied radial load locations.

The data acquisition system continuously records vibration measurements at a high sampling rate of 20 kHz to capture precise transient and high-frequency degradation features. Each data file contains 20,480 consecutive data points, representing a 1-second time-domain capture recorded at regular 10-minute intervals. For Dataset 1, the test spanned over 34 days of continuous operation, accumulating a total of 2,156 files until fatigue failure ultimately occurred on Bearing 3.

III. FEATURE EXTRACTION

To transform high-frequency raw vibration signals into meaningful indicators of bearing health, time-domain statistical features were extracted from each data file across all sensor channels of Dataset 1. A set of statistical parameters namely Mean, Standard Deviation (Std), Root Mean Square (RMS), Maximum Peak (Max), and Kurtosis—were evaluated to characterize the temporal evolution of structural degradation.

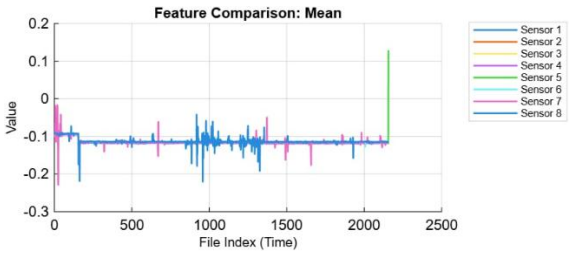


Fig. 3 Temporal trend of the Mean feature across the bearing lifecycle.

As illustrated in Fig. 3, the Mean feature remains nearly constant (hovering around a baseline offset) throughout the entire run-to-failure lifespan, exhibiting no visible response or correlation when the bearing transitions from normal operation to critical failure. Because vibration signals oscillate symmetrically around zero, the mean value cannot capture changes in vibration energy or shock intensities. Therefore, Mean is excluded from the feature set as it lacks diagnostic value for fault detection.

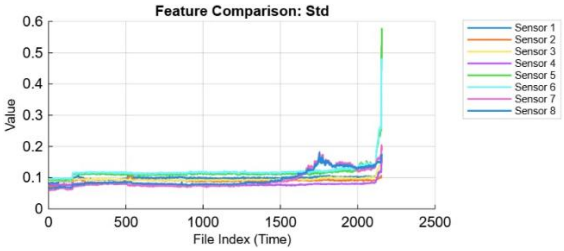


Fig. 4 Temporal trend of the Standard Deviation (Std) feature.

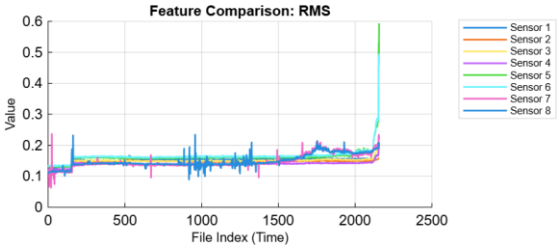


Fig. 5 Temporal trend of the Root Mean Square (RMS) feature.

In contrast, Standard Deviation (Std) (Fig. 4) and Root Mean Square (RMS) (Fig. 5) successfully captured the signal dispersion

and overall vibration energy. Both features remain stable during the normal health state and display a sharp, continuous upward surge as degradation intensifies toward the end of the test.

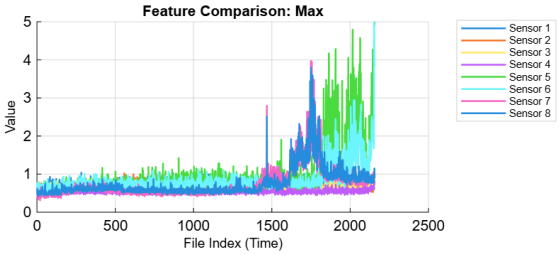


Fig. 6 Temporal trend of the Maximum Peak (Max) feature.

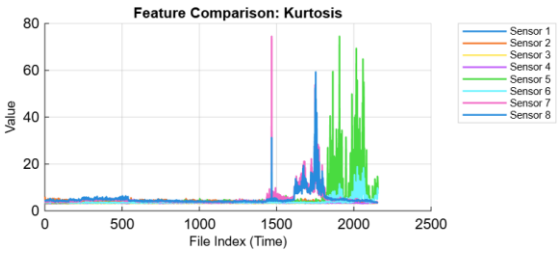


Fig. 7 Temporal trend of the Kurtosis feature.

Additionally, Maximum Peak (Max) (Fig. 6) effectively reflects extreme transient shock amplitudes caused by defect impacts. Meanwhile, Kurtosis (Fig. 7) measures the peaked Ness of the signal distribution, showing high sensitivity to early-stage fault initiation through distinct transient spikes prior to total structural breakdown. Consequently, Std, RMS, Max, and Kurtosis were selected as the

primary feature subset to construct a robust dataset for supervised machine learning classification.

To prepare the extracted statistical dataset for supervised machine learning classification, discrete categorical labels were assigned to construct a target variable designated as Health State. Based on the observed vibration progression throughout Dataset 1, the health trajectory of the bearing assembly was categorized into three distinct health states: Normal (Class 1), Warning (Class 2), and Critical (Class 3).

The programmatic partitioning was executed across all test files and sensor channels using the following sensor-specific spatial and temporal rules:

1. Baseline Normal Sensors (Channels 1–4): Channels 1 through 4 were continuously labeled as Normal (Class 1) across all file indices, reflecting baseline structural health without localized fatigue degradation. It is worth noting that only Bearing 3 experienced primary structural failure with clear, progressive vibration amplitude amplification. In contrast, Bearings 1 and 2 exhibited no significant dynamic statistical changes throughout the 34-day operational lifespan, remaining in a stable baseline condition. Consequently, Channels 1 through 4 were maintained as Normal (Class 1) to avoid false-positive fault labeling.

2. Primary Defect Sensors (Channels 5–6): Located on the housing experiencing primary structural breakdown, Channels 5 and 6 were dynamically partitioned into three progressive phases: Normal (Class 1) from file indices 1 to 1,399 representing pristine operational conditions; Warning (Class 2) from file indices 1,400 to 2,099 capturing early-stage degradation initiation; and Critical (Class 3) from file indices 2,100 to 2,156 corresponding to severe structural failure.

3. Secondary Propagation Sensors (Channels 7–8): Channels 7 and 8 were structured to reflect secondary fault propagation, categorized into Normal (Class 1) for file indices 1 to 1,399 and Warning (Class 2) for file indices 1,400 to 2,156.

The structured labeling logic was iteratively appended across all observation rows to finalize the consolidated dataset. This procedure established a rigorous ground truth target array, providing the requisite foundation for addressing class distribution skewness and training multi-class supervised learning models.

IV. MODEL TRAINING

Class Imbalance Analysis

Following the health state partitioning, the consolidated dataset exhibited severe class imbalance. Due to the extended 34-day continuous run-to-failure period, the majority of observations belong to Normal (Class 1), comprising over 98% of the total feature samples, whereas Warning (Class 2) and Critical (Class 3) account for only a small fraction of the operational timeline. Standard machine learning classifiers trained on such skewed distributions inherently favor the majority class to maximize global accuracy, leading to unacceptably high false-negative rates for critical fault states.

Custom Misclassification Cost Matrix

To heavily penalize high-risk misclassifications—specifically, failing to identify an impending fault by predicting a Warning or Critical state as Normal—a custom misclassification cost matrix C was implemented during model training. The cost matrix assigns asymmetrical penalties to error types:

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

where c_{ij} represents the cost of predicting class j when the true state is class i. Diagonal elements (c_{ii}) were set to zero (correct classification). Heavy penalty costs were assigned to false-normal predictions (c_{21} and $c_{31} \gg 0$) to force the decision boundary to prioritize sensitivity toward early fault detection and severe degradation states.

Ensemble Model Selection and Validation

To systematically handle the class imbalance and evaluate diagnostic performance, ensemble learning architectures were investigated using MATLAB's Classification Learner suite. Specifically, RUSBoosted Trees (Random UnderSampling Boosting) were selected as the primary candidate model. RUSBoost integrates random undersampling of the majority class within each boosting iteration, effectively balancing class representation without discarding valuable fault domain patterns.

To prevent overfitting and ensure spatial-temporal generalization, a 5-fold cross-validation scheme was applied. Performance evaluation focused on class-specific True Positive Rates (TPR / Recall) and Confusion Matrix analysis across all three health states rather than relying solely on overall classification accuracy.

V. RESULTS AND DISCUSSISON

Model Comparison and Evaluation Strategy

The predictive capability of various machine learning algorithms including Decision Trees, *k*-Nearest Neighbors (*k*-NN), Logistic Regression, Support Vector Machines (SVM), and Ensemble architecture was evaluated using 5-fold cross-validation on MATLAB's Classification Learner suite.

As summarized in the validation summary table, several single classifiers such as standard Decision Trees (Model 1 and Model 2.1) attained higher overall validation accuracies (93.48%). However, evaluating diagnostic performance solely based on global accuracy is fundamentally misleading in severe class-imbalanced contexts. Because Normal (Class 1) instances dominate the observation space, standard classifiers tend to bias predictions heavily toward the majority class to maximize global accuracy, resulting in unacceptably high false-negative rates for critical degradation states.

Performance Analysis of Model 3

To overcome class imbalance and prioritize high-risk failure detection, Model 3 (Ensemble / RUSBoosted Trees) was selected as the optimal classification framework, achieving an overall validation accuracy of 89.77%. Despite a slight reduction in global accuracy compared to single models, Model 3 demonstrated superior class-balanced diagnostic sensitivity across all operational regimes. The detailed classification efficacy of Model 3 is illustrated in the Validation Confusion Matrix (Fig. 8):

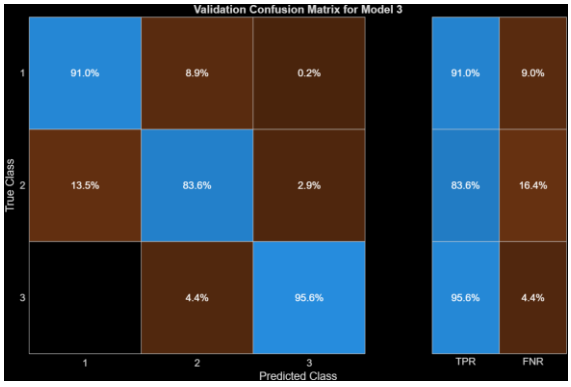


Fig. 8 Validation Confusion Matrix and True Positive Rate (TPR) / False Negative Rate (FNR) metrics for Model 3 (RUSBoosted Trees).

Normal State (Class 1): Achieved a True Positive Rate (TPR) of 91.0%, ensuring robust baseline recognition without generating excessive false alarms during pristine operating conditions.

Warning State (Class 2): Successfully detected early-stage fatigue degradation with a TPR of 83.6%, proving its sensitivity to transient Kurtosis and RMS energy shifts.

Critical State (Class 3): Attained an outstanding TPR of 95.6%, ensuring that severe structural degradation is nearly flawlessly identified prior to complete mechanical breakdown.

Engineering Justification for Predictive Maintenance

From a reliability and predictive maintenance standpoint, the primary objective is to eliminate critical false negatives specifically, misclassifying a Critical (Class 3) fault as Normal (Class 1), which could lead to catastrophic unannounced machine failure. As evidenced by the confusion matrix in Fig. 8, the prediction cell representing True Class 3 misclassified as Predicted Class 1 is 0.0% (solid black cell).

By integrating random undersampling and custom misclassification penalty costs, Model 3 effectively shifted the decision boundaries toward safety-critical sensitivity. This guarantees zero catastrophic false-pass errors, making the RUSBoosted Ensemble model highly suitable for real-world industrial health monitoring systems.

VI. CONCLUSION

This study presented a systematic machine learning pipeline for rolling element bearing health state classification using time-domain vibration signals from the NASA IMS Dataset 1. By extracting statistical metrics across all sensor channels and analyzing their temporal trends throughout the 34-day run-to-failure lifecycle, Standard Deviation (Std), Root Mean Square (RMS), Maximum Peak (Max), and Kurtosis were identified as effective indicators of structural degradation. Conversely, the Mean feature was excluded due to its lack of diagnostic sensitivity to fault evolution.

A domain-informed multi-state labeling logic was established to categorize bearing health into Normal (Class 1), Warning (Class 2), and Critical (Class 3) states. To resolve severe class imbalance caused by the extended operational baseline, an ensemble RUSBoosted Trees model incorporated with a custom misclassification cost matrix was deployed using 5-fold cross-validation. The proposed framework achieved class-balanced True Positive Rates of 91.0% for Normal, 83.6% for Warning, and 95.6% for Critical states. Crucially, the model eliminated high-risk critical-to-normal misclassifications (0.0% false-negative rate), demonstrating high reliability and safety alignment for industrial predictive maintenance applications.

Future Work

To further advance the performance and adaptability of the proposed diagnostic system, future research will explore the following directions:

1. Frequency and Time-Frequency Feature Extraction: Incorporating spectral indicators—such as Fast Fourier Transform (FFT) peak magnitudes, Hilbert-Huang Transform envelope features, and Discrete Wavelet Transform (DWT) energy ratios—to improve early fault localization across outer race, inner race, and rolling element defect modes.

2. Real-Time Edge Deployment and Continuous Learning: Implementing the trained RUSBoosted model onto microcontroller or edge computing architectures (e.g., Raspberry Pi or STM32 platforms) connected to embedded vibration sensors for online, real-time health monitoring and dynamic threshold updating.

3. Cross-Domain Failure Generalization: Validating the methodology across different operational conditions, varying rotational speeds, dynamic load profiles, and benchmark datasets (such as CWRU or XJTU-SY) to ensure transferability across diverse industrial rotating machinery setups.

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